

CHRISANTUS EZE

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EDUCATION

Oklahoma State University (OSU) | Ph.D., Computer Science (in-view) [January 2022 - In progress]

- **Research:** Foundation Models (LLMs/VLMs/VLAs/WMs), LoRA fine-tuning, multimodal reasoning, ROS, Mujoco, Pybullet, PyTorch, HuggingFace, LangChain, AWS, Docker, Python

Federal University of Technology, Owerri (FUTO), Nigeria | B.Eng. in Electrical & Electronic Eng October 2013 - October 2018

PUBLICATIONS

- **Eze, C., Julian, R., and Crick, C.** "Learning Object-Centric Spatial Reasoning for Sequential Manipulation in Cluttered Environments." [Under review] – Novel two-module architecture enabling robots to reason about occlusions and retrieve hidden objects through strategic sequential manipulation.
- **Eze, C. and Crick, C.** "Learning by Watching: A Review of Video-based Learning Approaches for Robot Manipulation." IEEE Access, 2025 – Comprehensive survey of state-of-the-art video learning methods for robotic systems.
- **Ezeakunne, U., Eze, C., and Liu, X.** "Data-Driven Fairness Generalization for Deepfake Detection." ICAART 2025 – Fairness-aware deepfake detection addressing demographic bias in generative models.
- **Eze, C. and Crick, C.** "A3: Active Adversarial Alignment for Source-Free Domain Adaptation." IEEE ICMLA 2024 – Domain adaptation framework for transferring pretrained policies to novel environments without target labels.
- **Eze, C. and Crick, C.** "Enhancing Human-Robot Collaboration by Exploring Intuitive Augmented Reality Design Representations." ACM/IEEE HRI 2023 – AR-based interface design for improved robot telepresence and collaboration.

RELEVANT EXPERIENCE

Graduate Researcher – Machine Learning & AI | **Department of Computer Science, OSU** January 2022 - Present

- Lead research on multimodal foundation models for long-horizon spatial reasoning and robotic manipulation.
- Developed **Unveiler**, a planning-control framework for sequential object retrieval in clutter (94% sim-to-real success).
- Built VLM-RAG pipelines for object grounding (+34% improvement); developed a source-free adaptation method (A3), improving cross-domain performance by 15%.
- Deployed and fine-tuned instruction-tuned VLMs using LoRA with PEFT, gradient checkpointing, and mixed-precision training, enabling GPU-efficient experimentation and private model adaptation.
- Developing logic-augmented spatial reasoning for VLMs via Answer Set Programming (ASP), benchmarking zero-shot performance, and fine-tuning on curated spatial datasets to improve compositional and relational reasoning.
- Designing and training pixel-space world models (WMs) and latent dynamics models from scratch; fine-tuning on task-specific manipulation data to improve long-horizon planning and spatial reasoning for vision-language-action (VLA) policies.
- Published 5 peer-reviewed papers in ML/robotics venues.

Engineering Intern – Applied ML for Process Systems | **Fractionation Research Inc.** June 2025 - August 2025

- Architected a real-time anomaly detection system for distributed control (50+ sensor streams), reducing false positives by 78% and contributing to 23% reduction in unplanned downtime.

Graduate Student Mentee – Applied ML Research | **Google** February 2023 - June 2025

- Co-developed sequential manipulation policies using visual reasoning, behavior cloning, and reinforcement learning (98% benchmark success).
- Contributed perception module integrating depth-based occlusion reasoning and learned affordances.

NON-RESEARCH PROJECTS

LLM-Powered Research Assistant (Lab-Mate)

- Built a GPT-4o Slack bot to auto-summarize and extract action items from over 100 research meetings.
- Built a RAG pipeline using semantic chunking and vector search; boosted contextual retrieval of lab protocols and discussions to 89% relevance.

VidSearch (Multimodal RAG System)

- Developed a video-grounded QA system with semantic chunking and vector retrieval;
- Used Pinecone for vector storage of transcript embeddings to ground LLM responses in specific video segments.